Research Journal

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September

Sep. 16, 2025 — Tuesday

- Revisited the credentials provided by the HPC facility at IUCAA last year – they are working fine
- Here are the current credentials:
  - **VPN**
    - Server: `moon.iucaa.in:4433`
    - Username: `parag.bhattacharya`
    - Password: `P@^@GBvpn2*2486`
    - Domain: `LocalDomain`
  - **ASSC account**
    - Command: `ssh -X paragb@192.168.11.34`
    - Password: *same as Win11*
  - **Using the installed software**

    ```
    source xraysoftinit
    heainit
    sas21init
    sas21
    ```

  - **Using Python3.10 etc.**

    ```
    module switch python/python310
    ```

  - **LDAP account credentials (at IUCAA)**
    - Username: `parag`
    - Password: `Saturn$890`

Sep. 17, 2025 — Wednesday

- Hopefully I will be able to do some polarimetric study of science data from IXPE, as proposed by Dr. Monmoyuri Baruah yesterday.
- The target for doing some work is 6 – 8 October 2025, during NEMA-XI at RGU, Arunachal Pradesh.
- I have two documents that I need to study:
  1. Bobrikova *et al.* (2024) – sent by Dr. Monmoyuri Baruah
  2. IXPE Quick Start Data Analysis Guide – downloaded from HEASARC
- To get started with the data analysis, I want to download the IXPE L2 data for two sources:
  1. GX 13+1 – a weakly magnetized neutron star (WMNS) studied by Bobrikova *et al.*
  2. Mrk 501 – a blazar which is provided as an example in the IXPE Quick Start Guide
- The science data are to be downloaded from the legacy HEASARC browse window as per the following parameters:
  - Object Name or Coordinates: `Mrk 501`
    Mission: IXPE
    Observation window: `July 2022`
  - Object Name or Coordinates: `GX 13+1`
    Mission: IXPE
    Observation window: `October 2023`

  Note: The current Xamin browse portal doesn't seem to load

Sep. 21, 2025 — Sunday

- Downloaded the L2 event files for the following dataset:
  1. `Mrk 501`
  2. `GX 13+1`
- Commands for file transfer using SCP – from Contabo Blog
- Commands for Mrk 501
  - Uploaded Mrk 501 `.tar` file from PC to ASSC server:

```
scp E:\Research\x-ray_polarimetry\data\raw\mrk-501.tar
 paragb@192.168.11.34:/home/paragb/xray-pol/dataset
```

- Navigated to data folder:

```
cd /home/paragb/xray-pol/dataset
```

- Extracted data from the `.tar` file

```
tar -xvf mrk-501.tar
```

- The extracted L2 files are contained in the folder: `/home/paragb/xray-pol/dataset/01004701/event_l2`
- The data folder has been accordingly renamed as: `mv -v 01004701/ mrk-501/`
- So the extracted L2 files are now contained in the folder: `/home/paragb/xray-pol/dataset/mrk-501/event_l2`
- The `.gz` files are unzipped as: `gunzip -v *.gz`

Sep. 22, 2025 — Monday

- Downloaded the relevant CALDB files for IXPE to the ASSC server using the commands

```
export CALDB=/home/paragb/caldb/
cd $CALDB
wget -v https://heasarc.gsfc.nasa.gov/FTP/caldb/data/
            ixpe/gpd/goodfiles_ixpe_gpd_20250225.tar.gz
wget -v https://heasarc.gsfc.nasa.gov/FTP/caldb/data/
            ixpe/xrt/goodfiles_ixpe_xrt_20241028.tar.gz
gunzip -v goodfiles_*.gz
tar -xvf goodfiles_*.tar
rm -rv goodfiles_*.tar
```

- Downloaded the data `.tar` files for Mrk 501 directly to the ASSC server.

Sep. 23, 2025 — Tuesday

- Began download of Heasoft v6.35.2 from remote server using the following command after opening the terminal in the desired folder.

```
wget -v https://heasarc.gsfc.nasa.gov/FTP/
    software/lheasoft/lheasoft6.35.2/heasoft-6.35.2src.tar.gz
```

- Installed MiKTeX using the following commands (MiKTeX webpage):

```
curl -fsSL https://miktex.org/download/key |
    sudo gpg --dearmor -o /usr/share/keyrings/miktex.gpg
echo "deb [signed-by=/usr/share/keyrings/miktex.gpg]
    https://miktex.org/download/ubuntu jammy universe" |
    sudo tee /etc/apt/sources.list.d/miktex.list
sudo apt-get update
sudo apt-get install miktex
miktexsetup finish
initexmf --set-config-value [MPM]AutoInstall=1
```

- Installed TeXstudio using the following commands:

```
sudo add-apt-repository ppa:sunderme/texstudio
sudo apt update
sudo apt install texstudio
```

- The `.tar` data files have been downloaded into the ASSC server at

```
/home/paragb/xray-pol/dataset/mrk-501/01004701/
```

- All `.gz` files have been extracted in the following folders:

```
auxil    event_l1    event_l2    hk
```

- To view the images on DS9: `ds9 ixpe01004701_det1_evt2_v01.fits &`
  - Enhancement of the image:
    `scale` → `log`
    `color` → `heat`
  - Creation of a source region:
    In the menu: e.g. `Region` → `Shape` → `Circle`
    `edit` → `region` → click + drag to draw the region
    Double-click on region to open the region dialog. Center in `fk5`, radius in `arcsec`.
  - Creation of a background region:
    In the menu: e.g. `Region` → `Shape` → `Annulus`
    `edit` → `region` → click + drag to draw the region
    Double-click on region to open the region dialog. Center in `fk5`, radius in `arcsec`.

- The region files are created in the L2 event folder. These files contain the following:
  - Inside `src.reg`:

    ```
    circle(16:53:51.766,+39:45:44.41,60.0")
    ```

  - Inside `bkg.reg`:

    ```
    circle(16:53:51.766,+39:45:44.41,252.0")
    -circle(16:53:51.766,+39:45:44.41,132.0")
    ```

- Extraction of spectra
  - Initializing `xselect`

    ```
    xselect prefix=mrk501
    xsel> read event "ixpe01004701_det1_evt2_v01.fits"
    ```

  - Extracting the source spectra (e.g. detector 1):

    ```
    xsel> filter region "src.reg"
    xsel> extract SPEC stokes=NEFF
    xsel> save spec mrk-501_d01_src_
    ```

    This creates 3 source spectra files, namely
    `mrk-501_d01_src_I.pha`
    `mrk-501_d01_src_Q.pha`
    `mrk-501_d01_src_U.pha`
  - Extracting the background spectra (e.g. detector 1):

    ```
    xsel> clear region
    xsel> filter region "bkg.reg"
    xsel> extract SPEC stokes=NEFF
    xsel> save spec mrk-501_d01_bkg_
    ```

    This creates 3 background spectra files, namely
    `mrk-501_d01_bkg_I.pha`
    `mrk-501_d01_bkg_Q.pha`
    `mrk-501_d01_bkg_U.pha`
  - Finding the appropriate RMF files:
    1. Find the observation in the `ixmaster` catalogue using the HEASARC search form
    2. Copy the entry for the object of interest under the `time` column, e.g. `2022-07-09 23:23:24.082` for the Mrk 501 obs. ID 01004701

3. Create a CALDB variable that points to the folder containing the IXPE calibration database as
   ```
   export CALDB=/home/paragb/caldb
   ```

4. Use the `time` value in the `quzcif` command, for detector unit 1 (DU1) of the gas pixel detector instrument (GPD), as follows:
   ```
   quzcif IXPE GPD DU1 - MATRIX 2022-07-09 23:23:24.082 -
   ```

   This would give the following output:
   ```
   /home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
       ixpe_d1_20211209_01.rmf 1
   /home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
       ixpe_d1_20211209_alpha075_01.rmf 1
   ```

5. We have to use the file whose name contains the string `alpha075` when the events are weighted by either the `NEFF` or `SIMPLE` weighting schemes. In our case, we used `NEFF` while extracting the source and background spectra and so we can create a soft link to the `alpha075` file as follows:
   ```
   ln -s /home/paragb/caldb/data/ixpe/gpd/cpf/rmf/
       ixpe_d1_20211209_alpha075_01.rmf mrk-501_d01_rmf.fits
   ```

   This creates a symbolic link file named `mrk-501_d01_rmf.fits` which can be used as the RMF file in XSPEC later.

- Creating the MRF (for Q and U spectra) and ARF (for I spectrum) files:

  1. Sent the following error report to NASA Helpdesk:
     ```
     Recently I have been introduced to IXPE science data analysis
     and I have been trying to replicate the results of the
     walkthrough in section 5 of the IXPE quick start analysis guide.

     I have been able to successfully generate the I, Q and U
     spectra as described in the aforementioned document.
     However, I am running into an error when I try to create
     the MRF file for the Q and U spectra.

     I have copied the attitude file from the /hk folder into
     the /event_l2 folder. The input and output of the console
     are as follows:
     [paragb@localhost event_l2]$ ixpecalcarf \
     > evtfile=ixpe01004701_det1_evt2_v01.fits.gz \
     > attfiles=ixpe01004701_det1_att_v01.fits.gz \
     > arfout=mrk-501_d01.mrf \
     ```

```
> specfile=none radius=1.0 weight=1 resptype=mrf
Full path name of a Level 2 attitude file.
        [ixpe01004701_det1_att_v01.fits]
        attfiles=ixpe01004701_det1_att_v01.fits.gz
[Errno 2] No such file or directory: 'none'

The issue persists even when I provide the full path name
of the attitude file:
[paragb@localhost event_l2]$ ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits.gz \
> attfiles=/home/paragb/xray-pol/dataset/mrk-501/
    01004701/event_l2/ixpe01004701_det1_att_v01.fits.gz \
> arfout=mrk-501_d01.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf
Full path name of a Level 2 attitude file.
    [attfiles=ixpe01004701_det1_att_v01.fits.gz]
        /home/paragb/xray-pol/dataset/mrk-501/01004701/event_l2/
        ixpe01004701_det1_att_v01.fits.gz
[Errno 2] No such file or directory: 'none'

I am bamboozled because I have followed the exact same steps
as given in the document. I am unable to figure out as
to where I have been going wrong.

I would be grateful to receive advice from you in this matter.
```

Sep. 24, 2025 — Wednesday

- Received the following response from NASA Helpdesk:

```
Hello,

We were unable to reproduce the error you reported.
To help resolve this issue, please try the following:

1. Execute 'punlearn ixpecalcarf' to clean your parameter
file, as we suspect there may be a stale parameter
value causing the problem.

2. Please also check your attitude file configuration,
as your output shows the attitude file name appearing
inconsistently with and without the .gz extension.
```

```
Please let us know if this resolves the issue or if
you continue to experience problems.

Thanks,
Mihoko Yukita
```

- Installed Heasoft v6.35.2 on laptop.

Sep. 25, 2025 — Thursday

- Started analysis of data for Mrk 501 locally on my laptop.
- Installed Heasoft 6.35.2 locally on my PC (running on Ubuntu 22.04 LTS).
- Remote access for CALDB set up in the folder:
  `/home/pararover/Software/astro-soft/caldb`
  which contains the files `alias_config.fits` and `caldb.config`
- The `.bashrc` file is appended with the following lines:

```
CALDBCONFIG=/home/pararover/Software/astro-soft/caldb/
    caldb.config; export CALDBCONFIG
CALDBALIAS=/home/pararover/Software/astro-soft/caldb/
    alias_config.fits; export CALDBALIAS
CALDB=https://heasarc.gsfc.nasa.gov/FTP/caldb; export CALDB
```

- Data from HEASARC are downloaded and extracted into the following folder:

```
/home/pararover/Documents/Research/xray-pol/dataset/mrk-501/01004701
```

Sep. 28, 2025 — Sunday

- Downloaded `.tar` files for IXPE observations GX 13+1, with the Obs. ID. 02006801, in two batches because HEASARC does not allow downloads more than 2 GB.
  1. `event_l1/` for ~ 1.6 GB
  2. `event_l2/`, `auxil/` and `hk/` for ~ 700 MB
- The above two batches of `.tar` files need to be extracted and combined together into a single folder named `gx13+1/` with the following folder hierarchy:
  - `gx13+1/`
    * `02006801`

- auxil/
- event_l1/
- event_l2/
- hk/

October

Oct. 01, 2025 — Wednesday

- Found this good paper on polarized X-ray emission from a BH XRB named 4U 1630-47
  https://iopscience.iop.org/article/10.3847/2041-8213/acd77b

- For their analysis Rawat *et al.* have used IXPEOBSSIM
  https://ixpeobssim.readthedocs.io/en/latest/index.html

- I intend to explore IXPEOBSSIM further in this direction of research. To this end, I have installed the same on my PC using the command (after running `heainit`):
  ```
  pip3 install ixpeobssim
  ```

- IXPEOBSSIM requires the following Python packages to run:
  `numpy` `scipy` `matplotlib` `astropy` `regions` `skyfield`

- I have upgrade all of the above packages individually by running the command:
  ```
  pip3 install <<package-name>> --upgrade
  ```

- Before proceeding, I need to look into the tutorials provided here:
  https://ixpeobssim.readthedocs.io/en/latest/tutorial.html

- Logs from my tryout of the tutorial:

  ```
  >>> import numpy
  >>>
  >>> from ixpeobssim.config import file_path_to_model_name
  >>> About to create folder /home/pararover/ixpeobssimdata...
  >>> Folder succesfully created.
  >>> About to create folder /home/pararover/ixpeobssimdata/benchmarks...
  >>> Folder succesfully created.
  >>> About to create folder /home/pararover/ixpeobssimauxfiles...
  >>> Folder succesfully created.
  Matplotlib is building the font cache; this may take a moment.
  /home/pararover/.local/lib/python3.10/site-packages/matplotlib/
      projections/__init__.py:63:
      UserWarning: Unable to import Axes3D. This may be due to
      multiple versions of Matplotlib being installed
  ```

```
    (e.g. as a system package and as a pip package). As a result,
    the 3D projection is not available.
  warnings.warn("Unable to import Axes3D.
    This may be due to multiple versions of "
>>> from ixpeobssim.srcmodel.roi import xPointSource, xROIModel
>>> from ixpeobssim.srcmodel.spectrum import power_law
>>> from ixpeobssim.srcmodel.polarization import constant
>>>
```

Oct. 16, 2025 — Thursday

- The solution provided by the IXPE Helpdesk is working.
- Was able to run `ixpecalcarf` to generate MRF files for D01, D02 and D03.
- However, the following warning appeared:

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf
/home/pararover/.local/lib/python3.10/site-packages/matplotlib/
projections/__init__.py:63: UserWarning: Unable to import Axes3D.
This may be due to multiple versions of Matplotlib being
installed (e.g. as a system package and as a pip package). As a
result, the 3D projection is not available.
  warnings.warn("Unable to import Axes3D. This may be due to
  multiple versions of "
```

- The `ixpecalarf` task has been run as follows:

  1. For detector #1:
     To generate an MRF (for Q and U spectra):

     ```
     pararover@shadowfax: ixpecalcarf \
     > evtfile=ixpe01004701_det1_evt2_v01.fits \
     > attfiles=ixpe01004701_det1_att_v01.fits \
     > arfout=mrk-501_d01_QU.mrf \
     > specfile=none radius=1.0 weight=1 resptype=mrf
     ```

     To generate an ARF (for I spectrum):

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det1_evt2_v01.fits \
> attfiles=ixpe01004701_det1_att_v01.fits \
> arfout=mrk-501_d01_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf
```

2. For detector #2:
   To generate an MRF (for Q and U spectra):

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02_QU.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf
```

   To generate an ARF (for I spectrum):

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det2_evt2_v01.fits \
> attfiles=ixpe01004701_det2_att_v01.fits \
> arfout=mrk-501_d02_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf
```

3. For detector #3:
   To generate an MRF (for Q and U spectra):

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det3_evt2_v01.fits \
> attfiles=ixpe01004701_det3_att_v01.fits \
> arfout=mrk-501_d03_QU.mrf \
> specfile=none radius=1.0 weight=1 resptype=mrf
```

   To generate an ARF (for I spectrum):

```
pararover@shadowfax: ixpecalcarf \
> evtfile=ixpe01004701_det3_evt2_v01.fits \
> attfiles=ixpe01004701_det3_att_v01.fits \
> arfout=mrk-501_d03_I.arf \
> specfile=none radius=1.0 weight=1 resptype=arf
```

- The list of necessary files for detector #1 are:

```
mrk-501_d01_src_I.pha
```

```
mrk-501_d01_src_Q.pha
mrk-501_d01_src_U.pha
mrk-501_d01_bkg_I.pha
mrk-501_d01_bkg_Q.pha
mrk-501_d01_bkg_U.pha
mrk-501_d01.rmf -> ixpe_d1_20211209_alpha075_01.rmf
mrk-501_d01_I.arf
mrk-501_d01_QU.mrf
```

- The list of necessary files for detector #2 are:

```
mrk-501_d02_src_I.pha
mrk-501_d02_src_Q.pha
mrk-501_d02_src_U.pha
mrk-501_d02_bkg_I.pha
mrk-501_d02_bkg_Q.pha
mrk-501_d02_bkg_U.pha
mrk-501_d02.rmf -> ixpe_d2_20211209_alpha075_01.rmf
mrk-501_d02_I.arf
mrk-501_d02_QU.mrf
```

- The list of necessary files for detector #3 are:

```
mrk-501_d03_src_I.pha
mrk-501_d03_src_Q.pha
mrk-501_d03_src_U.pha
mrk-501_d03_bkg_I.pha
mrk-501_d03_bkg_Q.pha
mrk-501_d03_bkg_U.pha
mrk-501_d03.rmf -> ixpe_d2_20211209_alpha075_01.rmf
mrk-501_d03_I.arf
mrk-501_d03_QU.mrf
```

- Commands for spectrum file grouping of detector #1:

  1. For I spectrum:

```
fthedit mrk-501_d01_src_I.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d01_bkg_I.pha'" longstring=YES

fthedit mrk-501_d01_src_I.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d01.rmf'" longstring=YES
```

```
fthedit mrk-501_d01_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_I.arf'" longstring=YES
```

2. For Q spectrum:

```
fthedit mrk-501_d01_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d01_bkg_Q.pha'" longstring=YES

fthedit mrk-501_d01_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d01.rmf'" longstring=YES

fthedit mrk-501_d01_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_QU.mrf'" longstring=YES
```

3. For U spectrum:

```
fthedit mrk-501_d01_src_U.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d01_bkg_U.pha'" longstring=YES

fthedit mrk-501_d01_src_U.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d01.rmf'" longstring=YES

fthedit mrk-501_d01_src_U.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d01_QU.mrf'" longstring=YES
```

- Commands for spectrum file grouping of detector #2:

1. For I spectrum:

```
fthedit mrk-501_d02_src_I.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_I.pha'" longstring=YES

fthedit mrk-501_d02_src_I.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmf'" longstring=YES

fthedit mrk-501_d02_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_I.arf'" longstring=YES
```

2. For Q spectrum:

```
fthedit mrk-501_d02_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_Q.pha'" longstring=YES
```

```
fthedit mrk-501_d02_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmf'" longstring=YES

fthedit mrk-501_d02_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_QU.mrf'" longstring=YES
```

3. For U spectrum:

```
fthedit mrk-501_d02_src_U.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d02_bkg_U.pha'" longstring=YES

fthedit mrk-501_d02_src_U.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d02.rmf'" longstring=YES

fthedit mrk-501_d02_src_U.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d02_QU.mrf'" longstring=YES
```

- Commands for spectrum file grouping of detector #3:

1. For I spectrum:

```
fthedit mrk-501_d03_src_I.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d03_bkg_I.pha'" longstring=YES

fthedit mrk-501_d03_src_I.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d03.rmf'" longstring=YES

fthedit mrk-501_d03_src_I.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d03_I.arf'" longstring=YES
```

2. For Q spectrum:

```
fthedit mrk-501_d03_src_Q.pha keyword=BACKFILE operation=add \
value=" 'mrk-501_d03_bkg_Q.pha'" longstring=YES

fthedit mrk-501_d03_src_Q.pha keyword=RESPFILE operation=add \
value=" 'mrk-501_d03.rmf'" longstring=YES

fthedit mrk-501_d03_src_Q.pha keyword=ANCRFILE operation=add \
value=" 'mrk-501_d03_QU.mrf'" longstring=YES
```

3. For U spectrum:

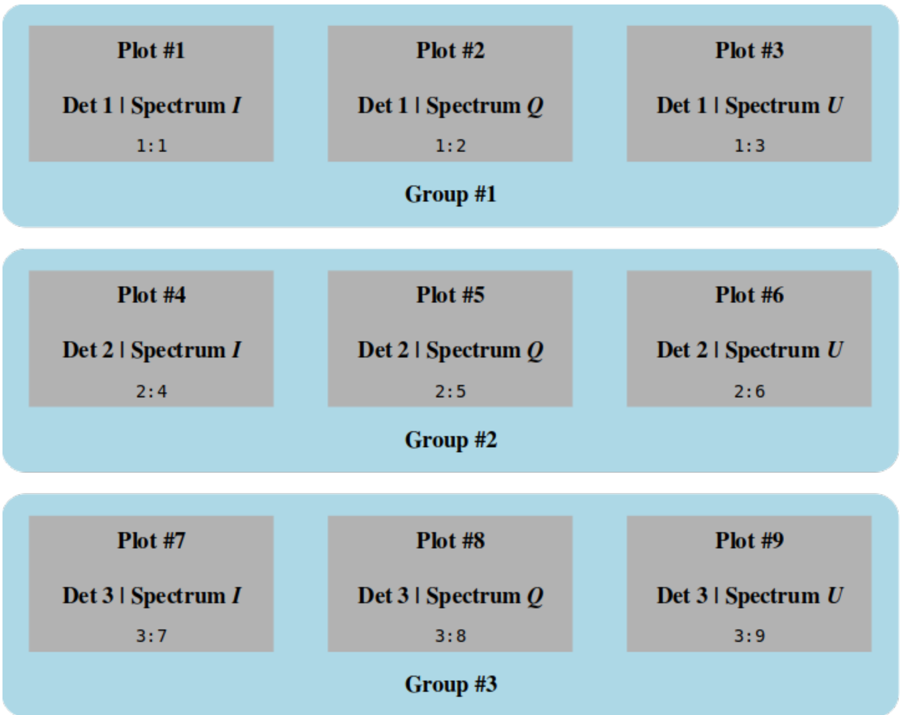
```
fthedit mrk-501_d03_src_U.pha keyword=BACKFILE operation=add \
value=" ’mrk-501_d03_bkg_U.pha’" longstring=YES

fthedit mrk-501_d03_src_U.pha keyword=RESPFILE operation=add \
value=" ’mrk-501_d03.rmf’" longstring=YES

fthedit mrk-501_d03_src_U.pha keyword=ANCRFILE operation=add \
value=" ’mrk-501_d03_QU.mrf’" longstring=YES
```

Oct. 17, 2025 — Friday

- Spectral fitting of all data groups to be performed as per the Quick Start Manual.
- Understanding the data and plot groups in XSPEC:



- The above data groups are created in XSPEC as follows:

```
data 1:1 mrk-501_d01_src_I.pha
data 1:2 mrk-501_d01_src_Q.pha
data 1:3 mrk-501_d01_src_U.pha

data 2:4 mrk-501_d02_src_I.pha
```

```
data 2:5 mrk-501_d02_src_Q.pha
data 2:6 mrk-501_d02_src_U.pha

data 3:7 mrk-501_d03_src_I.pha
data 3:8 mrk-501_d03_src_Q.pha
data 3:9 mrk-501_d03_src_U.pha
```

Oct. 18, 2025 — Saturday

- Steps that were used to analyze the spectra of Mrk 501 in XSPEC are given below:

  I. Reading the data:

  ```
  XSPEC12> data 1:1 mrk-501_d01_src_I.pha
  XSPEC12> data 1:2 mrk-501_d01_src_Q.pha
  XSPEC12> data 1:3 mrk-501_d01_src_U.pha
  XSPEC12> data 2:4 mrk-501_d02_src_I.pha
  XSPEC12> data 2:5 mrk-501_d02_src_Q.pha
  XSPEC12> data 2:6 mrk-501_d02_src_U.pha
  XSPEC12> data 3:7 mrk-501_d03_src_I.pha
  XSPEC12> data 3:8 mrk-501_d03_src_Q.pha
  XSPEC12> data 3:9 mrk-501_d03_src_U.pha
  ```

  II. Filtering the photons within an energy bandwidth:

  ```
  XSPEC12> ignore *:**-2.0 8.0-**
  ```

  Here the first `*` means to apply this filter on all data groups. Then `**-2.0 8.0-**` means to ignore all photon counts for $E < 2\,\mathrm{keV}$ and $E > 8\,\mathrm{keV}$, i.e. only the the $2\,\mathrm{keV} \leqslant E \leqslant 8\,\mathrm{keV}$ bandwidth is to be considered.

  III. Applying the composite model:

  ```
  XSPEC12> model constant*tbabs*(polconst*powerlaw)
  ```

  Here `constant` models an energy-independent multiplicative factor, `tbabs` models the intervening absorption, `polconst` models a constant polarization at all energies and `powerlaw` models a simple photon power law.

  IV. Parameter #1 is frozen at 1. Parameters 7 and 13 are untied and thawed. The parameter list prior to fitting looks as follows:

  ```
  XSPEC12> show par

  Parameters defined:
  ```

```
========================================================================
Model constant<1>*TBabs<2>(polconst<3>*powerlaw<4>) Source No.: 1
Active/On
Model Model Component  Parameter  Unit     Value
par  comp
Data group: 1
1     1   constant   factor              1.00000      frozen
2     2   TBabs      nH         10^22    1.00000      +/-  0.0
3     3   polconst   A                   1.00000      +/-  0.0
4     3   polconst   psi        deg      45.0000      +/-  0.0
5     4   powerlaw   PhoIndex            1.00000      +/-  0.0
6     4   powerlaw   norm                1.00000      +/-  0.0
Data group: 2
7     1   constant   factor              1.00000      +/-  0.0
8     2   TBabs      nH         10^22    1.00000      = p2
9     3   polconst   A                   1.00000      = p3
10    3   polconst   psi        deg      45.0000      = p4
11    4   powerlaw   PhoIndex            1.00000      = p5
12    4   powerlaw   norm                1.00000      = p6
Data group: 3
13    1   constant   factor              1.00000      +/-  0.0
14    2   TBabs      nH         10^22    1.00000      = p2
15    3   polconst   A                   1.00000      = p3
16    3   polconst   psi        deg      45.0000      = p4
17    4   powerlaw   PhoIndex            1.00000      = p5
18    4   powerlaw   norm                1.00000      = p6
________________________________________________________________________
```

V. Renormalizing and fitting the model to the spectrum gives the following parameter values:

```
========================================================================
Model constant<1>*TBabs<2>(polconst<3>*powerlaw<4>) Source No.: 1
Active/On
Model Model Component  Parameter  Unit     Value
par  comp
Data group: 1
1     1   constant   factor              1.00000      frozen
2     2   TBabs      nH         10^22    0.541250     +/-  0.117460
3     3   polconst   A                   7.25453E-02  +/-  1.90183E-02
4     3   polconst   psi        deg      -49.1296     +/-  7.50921
5     4   powerlaw   PhoIndex            2.60742      +/-  5.68093E-02
6     4   powerlaw   norm                9.63365E-02  +/-  8.13714E-03
Data group: 2
```

```
7     1   constant   factor             1.00375      +/-  9.94110E-03
8     2   TBabs      nH         10^22  0.541250      = p2
9     3   polconst   A                 7.25453E-02   = p3
10     3   polconst   psi       deg    -49.1296      = p4
11     4   powerlaw   PhoIndex         2.60742       = p5
12     4   powerlaw   norm             9.62997E-02   = p6
Data group: 3
13     1   constant   factor           0.924722      +/-  1.00195E-02
14     2   TBabs      nH        10^22  0.541250      = p2
15     3   polconst   A                7.25453E-02   = p3
16     3   polconst   psi       deg    -49.1296      = p4
17     4   powerlaw   PhoIndex         2.60742       = p5
18     4   powerlaw   norm             9.63365E-02   = p6
_________________________________________________________________


Fit statistic: Chi-Squared 131.57 using 149 bins,
          spectrum 1, group 1.
Chi-Squared      159.05     using 149 bins, spectrum 2, group 1.
Chi-Squared      122.69     using 149 bins, spectrum 3, group 1.
Chi-Squared      130.71     using 149 bins, spectrum 4, group 2.
Chi-Squared      118.58     using 149 bins, spectrum 5, group 2.
Chi-Squared      157.11     using 149 bins, spectrum 6, group 2.
Chi-Squared       60.87     using 149 bins, spectrum 7, group 3.
Chi-Squared       82.38     using 149 bins, spectrum 8, group 3.
Chi-Squared       69.58     using 149 bins, spectrum 9, group 3.
Total fit statistic                  1032.53     with 1334 d.o.f.

Test statistic : Chi-Squared        1032.53     using 1341 bins.
Null hypothesis probability of 1.00e+00 with 1334 degrees of freedom
```

VI. We are interested in finding the errors in the polarization degree A and polarization angle ψ, which are represented by the parameters A and `psi` in the `polconst` model component.

- To find the 99% confidence intervals for, say data group 1, we have

```
XSPEC12> error 6.635 3
XSPEC12> error 6.635 4
```

- To find the 1σ confidence intervals for, say data group 1, we have

```
XSPEC12> error 1.0 3
XSPEC12> error 1.0 4
```

Note: We cannot compute the errors in the same parameters for the other two data groups (i.e. 2 and 3) because these parameters are tied to their counterparts in data group 1. Any attempt to do so will be flagged as an error by XSPEC as follows:

```
XSPEC12> error 6.635 9
*** Parameter 9 is not a variable model parameter and
 no confidence range will be calculated.
Parameter   Confidence Range (6.635)
XSPEC12> error 6.635 10
*** Parameter 10 is not a variable model parameter and
 no confidence range will be calculated.
Parameter   Confidence Range (6.635)
XSPEC12> error 6.635 15
*** Parameter 15 is not a variable model parameter and
 no confidence range will be calculated.
Parameter   Confidence Range (6.635)
XSPEC12> error 6.635 16
*** Parameter 16 is not a variable model parameter and
 no confidence range will be calculated.
Parameter   Confidence Range (6.635)
```

VII. To visualize the contour plot, we need the following commands:

```
XSPEC12> steppar 3 0.00 0.21 50 4 -90 0 60
XSPEC12> plot contour ,,4 1.386, 4.61 9.21 13.81
```

The `steppar` command generates a 2D grid of A and ψ values with 50 steps in the range $0 < A < 0.21$ and 60 steps in the range $-90^\circ < \psi < 0^\circ$.
The `plot contour` command plots contours at 50% (`1.386`), 90% (`4.61`), 99% (`9.21`) and 99.9% (`13.81`) confidence levels for 2D errors in χ^2.

- To estimate the MDP$_{99}$ using PIMMS, we need to do the following:

1. Determine the model flux in the $2\,\mathrm{keV} \leqslant E \leqslant 8\,\mathrm{keV}$ bandwidth:

```
XSPEC12> flux 2.0 8.0
Spectrum Number: 1, 2, 3
Data Group Number: 1
Model Flux  0.015886 photons (8.7966e-11 ergs/cm^2/s) range
 (2.0000 - 8.0000 keV)

Spectrum Number: 4, 5, 6
Data Group Number: 2
Model Flux  0.015945 photons (8.8297e-11 ergs/cm^2/s) range
```

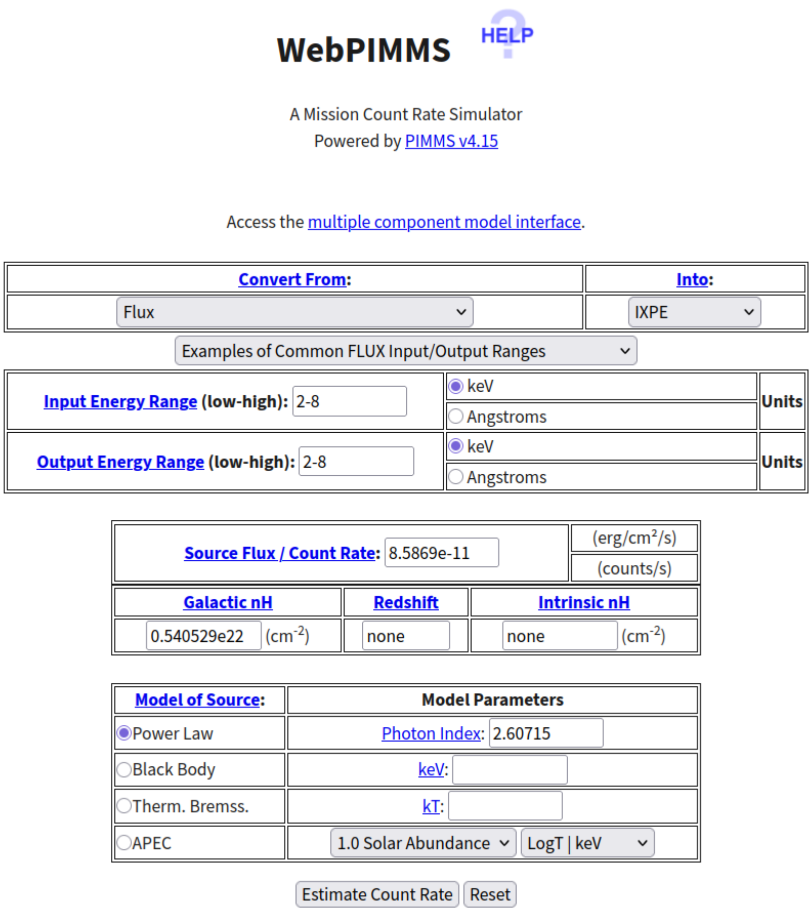
```
 (2.0000 - 8.0000 keV)

Spectrum Number: 7, 8, 9
Data Group Number: 3
Model Flux   0.01469 photons (8.1345e-11 ergs/cm^2/s) range
 (2.0000 - 8.0000 keV)
```

2. Collect the following values from the best-fit parameters so far:

$$n_H = 0.540\,529 \times 10^{22}\,\mathrm{cm}^{-2} \quad \text{(Galactic H column density)}$$
$$\Gamma = 2.60715 \quad \text{(photon index from powerlaw)}$$
$$\langle F \rangle = \left(\frac{8.7966 + 8.8297 + 8.1345}{3}\right) 10^{-11} = 8.5869 \times 10^{-11} \quad \text{(avg. flux from 3 detectors)}$$

3. Enter these values, along with some more, in WebPIMMS as shown:



4. WebPIMMS computes and displays the predicted count rates and MDP$_{99}$ values using different algorithms, some of which are shown below: We find,

IXPE OPEN Modulation Factor

Modulation Factor = **0.3140**
In 10000.0 s, MDP_99 = **16.13%**
In 100000.0 s, MDP_99 = **5.10%**

IXPE GRAY TR21D

The Total count rate in the selected band:
PIMMS predicts **1.817E-01** cps with IXPE GRAY TR21D

IXPE GRAY ER21D

The Effective count rate in the selected band:
PIMMS predicts **1.549E-01** cps with IXPE GRAY ER21D

IXPE GRAY MR21D

The Modulated count rate in the selected band:
PIMMS predicts **6.300E-02** cps with IXPE GRAY MR21D

IXPE GRAY Modulation Factor

Modulation Factor = **0.4069**
In 10000.0 s, MDP_99 = **26.80%**
In 100000.0 s, MDP_99 = **8.48%**

for instance, under 'IXPE OPEN Modulation Factor', that for an exposure time of $100\,\mathrm{ks}$, the MDP$_{99}$ is 5.15%.

$$T = 100\,\mathrm{ks} \implies \text{MDP}_{99} = 5.10\% = 0.051$$

Now MDP$_{99}$ scales with exposure time T as $\text{MDP}_{99} \propto \dfrac{1}{\sqrt{T}}$. So we have

$$\frac{\text{MDP}_{99}(\text{actual})}{\text{MDP}_{99}(\text{simulated})} = \sqrt{\frac{T(\text{simulated})}{T(\text{actual})}}$$

Running the `show data` command gives us the actual exposure time (e.g. for I spectrum by detector #1) to be $97800\,\mathrm{s} = 97\,\mathrm{ks}$.
Therefore, substituting $T(\text{simulated}) = 100\,\mathrm{ks}$, $T(\text{actual}) = 97\,\mathrm{ks}$ and $\text{MDP}_{99}(\text{simulated}) = 5.10\%$ above, we obtain

$$\text{MDP}_{99}(\text{actual}) = \sqrt{\frac{100}{97}}(5.10\%) = 5.18\% = 0.0518$$

This implies that, for an *unpolarized source* with the same physical parameters, an IXPE observation will return a value $A > 0.0518$ only 1% (=100%-99%, the 99 is from MDP$_{99}$) of the time.

- The chain of logic is as follows:

  1. Scale PIMMS number because the real observation time was a bit shorter, i.e. MDP$_{99} = 5.18\%$.
  2. MDP$_{99}$ means "only 1% chance a zero-polarisation source would look this big or bigger."
  3. Measured value of 7.24% (from best-fit) is above 5.18%, and the measurement's uncertainty of 1.9% is small (i.e. $5.18/1.9 \approx 3.8$).
  4. Therefore the measured polarisation is very unlikely to be a random fluke – it is a highly probable real detection.

2026

May

May. 17, 2026 — Sunday

- All data have been now migrated to the following location on my PC:

  `/home/pararover/AstroDA/xray-pol/dataset`

May. 27, 2026 — Wednesday

- Checked my VPN and ASSC account credentials at the HPC facility in IUCAA which have been lying dormant for quite some time – both are working fine.
- For GX13+1 I had downloaded the data corresponding to the ObsID 02006801.
- Extracted from the `.tar` file to find the following folder hierarchy:
  - `02006801`
    * `auxil/`
    * `event_l1/`
    * `event_l2/`
    * `hk/`
- We need to `gunzip` all the files in each of the sub-folders above.
- In addition, we need to copy the following files from the `hk/` folder and paste into the `event_l2/` folder:
  - `ixpe02006801_det1_att_v01.fits`
  - `ixpe02006801_det2_att_v01.fits`
  - `ixpe02006801_det3_att_v01.fits`
- As per section 2.1 in Bobrikova *et al.* (2024), the "source photons were selected in a circular region with a radius of 100" centered at the source position.".

- To obtain the source region, we need to open any of the L2 event files using DS9. For this, we have to navigate to the `event_l2/` folder and open any of the files from detector 1, 2 or 3 using DS9, i.e.

```
ds9 ixpe02006801_det1_evt2_v01.fits &
```

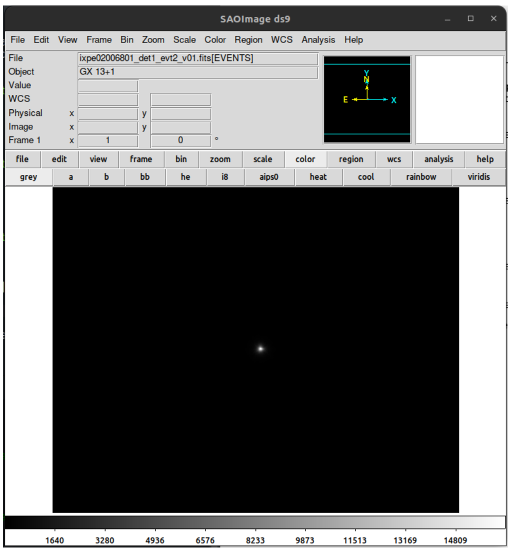
or

```
ds9 ixpe02006801_det2_evt2_v01.fits &
```

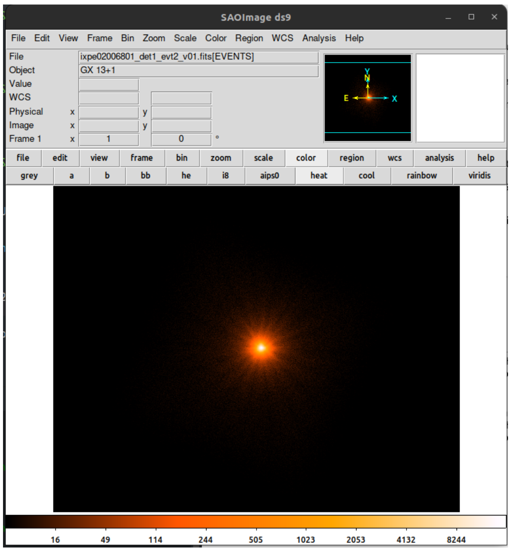
or

```
ds9 ixpe02006801_det3_evt2_v01.fits &
```

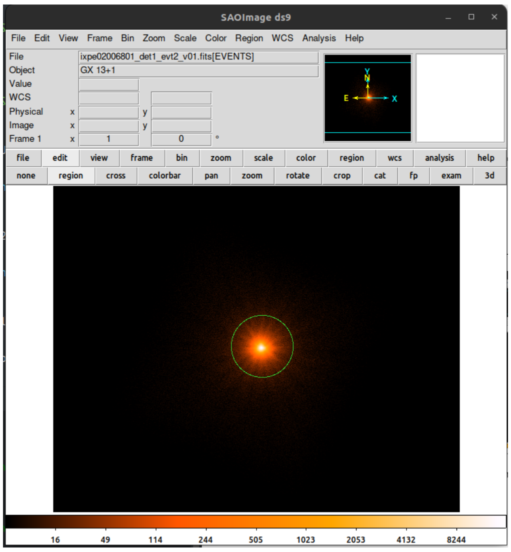
- Suppose we choose the file `ixpe02006801_det1_evt2_v01.fits`. Upon opening in DS9, we obtain as follows:



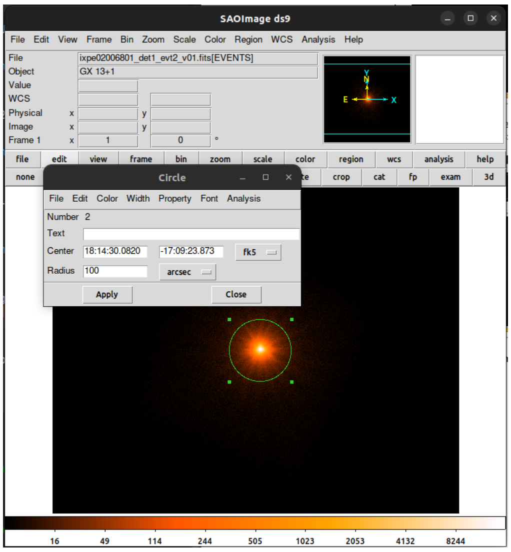
- Following the DS9 steps from Sep. 23, 2025 (Tuesday), on page 4, we obtain an enhancement of the image as follows:



- The source region is obtained by clicking on the centroid position and dragging to obtain a circle enclosing the source as follows:



- Double-clicking on the circular region opens the regions dialog. We have to ensure that the `Center` is in `fk5` and the `radius` is in `arcsec`. Then we type 100 in the `radius` textbox to ensure that the source region inside is of radius 100". Then click `Apply`, followed by `Close`.



- So we have to create a source region file named `src.reg` in the `event_l2/` folder. The file `src.reg` has the following contents copied and pasted from the regions dialog:

```
circle(18:14:30.0820,-17:09:23.873,100.0")
```

- GX13+1 is a bright source and so, as per section 2 of Di Marco *et al.* (2023), we don't need to subtract any background photons.

May. 31, 2026 — Sunday

Reducing IXPE science data for analysis

- Obtaining the attitude files
  1. Go to the `hk/` folder and copy-paste the following attitude files into the `event_l2/` folder

     `ixpe*_det?_att_v01.fits`

- Selecting the source and background regions
  1. Go to the `event_l2/` folder and open any of the L2 event files using DS9

     `ixpe*_det?_evt2_v01.fits`

  2. Draw a circular source region with centroid at the location of the source and with radius in arcsec as per literature.

3. Create a region file, such as `src.reg` and type the region info with coordinates in fk5 format and radius in arcseconds.

- Extracting the source and background photons
  1. We need to use xselect for this task.
  2. Make use of the steps mentioned in the journal entry dated 23 Sep. 2025 (Tuesday).
  3. This would create the I, Q and U spectra files for all 3 detectors, i.e. a total of 9 files.

- Obtain the relevant RMF file
  1. Obtain the observation using the HEASARC search form.
  2. Locate the row containing the `ObsID` of interest.
  3. Copy the entry under the `time` column for that row.
  4. Ensure that the CALDB variable points to the calibration database with the following command on the Terminal:

     ```
     echo $CALDB
     ```

     if everything is right, then we will get an output such as the following:

     ```
     https://heasarc.gsfc.nasa.gov/FTP/caldb
     ```

     which is the remote CALDB host at HEASARC.
  5. If the above does not appear, then set the CALDB variable with the following command:

     ```
     export CALDB=https://heasarc.gsfc.nasa.gov/FTP/caldb
     ```

     or add the above line in the `.bashrc` file and compile it.
  6. Use the time value obtained in step 3 above in the `quzcif` command. An example is as follows

     ```
     quzcif IXPE GPD DU1 - MATRIX 2023-10-17 16:41:45.850 -
     ```

     where GPD DU1 implies detector unit 1 of the gas pixel detector instrument. For detectors 2 and 3, we have to write DU2 and DU3 respectively, everything else remaining the same.
  7. After `quzcif` has completed its run, it would give a list of FTP locations of 2 files.

8. Download the file whose name contains the string '`alpha075`' using the `wget` command as follows:

   ```
   wget -v <<FTP URL>>
   ```

   For example:

   ```
   wget -v https://heasarc.gsfc.nasa.gov/FTP/caldb/data/ixpe/
   gpd/cpf/rmf/ixpe_d3_20211209_alpha075_01.rmf
   ```

9. Create soft links to these downloaded files, so that it is easier to refer to them, as follows

   ```
   ln -s <<file name>> <<link name>>
   ```

   For example:

   ```
   ln -s ixpe_d2_20211209_alpha075_01.rmf gx13_d02_rmf.fits
   ```

   following which we can use '`gx13_d02_rmf.fits`' as an alias for '`ixpe_d2_20211209_alpha075_01.rmf`'.

- Generate the ARF/MRF files

  1. We need to use `ixpecalcarf` for this task.
  2. A helpful trick is to list the relevant event file and attitude files first using the following commands. For example, in case of detector 1:

     ```
     ls det1_evt2
     ls det1_att
     ```

  3. Then run the `ixpecalcarf` as per steps mentioned in journal entry dated 16 Oct. 2025 (Thursday).

- Group the spectrum files

  1. We need to use `fthedit` for this task.
  2. Run the `fthedit` as per steps mentioned in journal entry dated 16 Oct. 2025 (Thursday).

June

Jun. 09, 2026 — Tuesday

- Found two nearby observations of GX 13+1 by IXPE and NuSTAR on the same day, i.e. 2024-02-25, as shown below

IXPE Master Catalog (ixmaster) Bulletin
Search radius used: 10.00'

Select All	Services	obsid	name	ra	dec	time	exposure 1 [s]	exposure 2 [s]	exposure 3 [s]	processing date	public date	obs type	issue flag	status	Search Offset [']
☐	O R N S D B	04002901	GX 13+1	18 14 31.6	-17 09 27	2025-09-08 01:47:59.800	147888.5019	148058.7024	148092.6483	2026-03-27 16:06:33.388	2025-09-18	proposed	0	archived	0.091 (GX 13+1)
☐	O R N S D B	03003401	GX 13+1	18 14 31.6	-17 09 27	2024-04-20 20:06:58.889	99886.2991	100011.1685	100112.6865	2024-04-29 18:25:24.441	2024-05-01	proposed	0	archived	0.091 (GX 13+1)
☐	O R N S D B	02006801	GX 13+1	18 14 31.4	-17 09 25	2023-10-17 16:41:45.850	98203.6295	98298.5532	98299.4520	2023-10-23 20:10:10.177	2023-10-31	proposed	0	archived	0.054 (GX 13+1)
☑	O R N S D B	03001101	GX 13+1	18 14 31.6	-17 09 27	2024-02-25 12:55:38.500	90670.4378	90818.5112	90799.4449	2024-02-29 21:39:02.413	2024-03-05	proposed	0	archived	0.091 (GX 13+1)

4 rows retrieved from ixmaster

NuSTAR Master Catalog (numaster) Bulletin
Search radius used: 12.00'

Select All	Services	name	ra	dec	time	obsid	status	exposure a [s]	observation mode	obs type	processing date	public date	issue flag	Search Offset [']
☑	O R N S D B	GX_13p1	18 14 41.5	-17 06 51	2024-02-25 12:56:10	30901010002	archived	32178	SCIENCE	N09	2024-02-27 02:24:00	2024-04-22	1	3.514 (GX 13+1)
☐	O R N S D B	GX_13p1	18 14 37.2	-17 08 12	2017-04-08 03:26:12	30301003002	archived	22711	SCIENCE	X16	2020-09-02 21:36:00	2018-05-08	0	1.845 (GX 13+1)
☐	O R N S D	GX13p1	18 14 20.2	-17 11 29	2025-08-30 12:31:06	91101330002	archived	20544	SCIENCE	TOO	2025-09-01 02:24:00	2025-09-05		3.384 (GX 13+1)
☐	O R N S D	GX_13p1	18 14 22.1	-17 11 46	2025-09-09 16:16:11	91101332002	archived	10579	SCIENCE	TOO	2025-09-11 12:00:00	2025-09-12	1	3.231 (GX 13+1)

4 rows retrieved from numaster

- Downloaded data for both observations.
- IXPE observation contained the same files and folders as before:
  `/auxil` `/event_l1` `/event_l2` `/hk`
- NuSTAR observation contained the following files and folders:
  `/auxil /event_cl /event_uf /hk nu30901010002.cat.gz pipe.log`

Jun. 15, 2026 — Monday

- Started the analysis of the 2024-02-25 data from IXPE.
- Providing a walkthrough of the steps below.

Background rejection filter

In Section 2.1 (IXPE), Bobrikova *et al.* (2024) state that they adopted the weighted analysis for their spectro-polarimetric study, following the baseline established by Di Marco *et al.* (2022a). This weighted scheme (using $w_k = \alpha^{0.75}$) is designed to improve polarimetric sensitivity by prioritizing well-reconstructed photoelectron tracks.

Bobrikova *et al.* (2024) also notes that they followed the recommendations of Di Marco *et al.* (2023b) regarding background handling. While they did not subtract the background due to the source's brightness, they did utilize the background rejection strategy described in that source to improve the quality of their data.

As per The Hitchhiker's Guide to IXPE Data Analysis and IXPE Quick Start Data Analysis Guide both, we can know if the background-rejection algorithm was applied in the pipeline by checking if the FITS header keyword XPFLGBGD=T has been included in the EVENTS extension of the L2 event file. We can do this in two different ways:

1. Using the `fkeyprint` command in the Terminal, for example as
   `fkeyprint ./event_l2/ixpe03001101_det1_evt2_v01.fits XPFLGBGD`
   The output did not return XPFLGBGD=T and hence the background rejection algorithm was not applied.

2. Using the `fv` command, we can click on the `Header` buttons, then manually search for XPFLGBGD=T in the Primary, EVENTS and GTI entension headers.

Hence, we have to manually perform the background rejection. Given below are the steps to do so:

- The Python script named `filter_background.py` is available on the GitHub page of IXPE-background.

- I have downloaded `filter_background.py` and placed it locally at `~/AstroDA/xray-pol/scripts/`

- Then I ran the script as follows:

```
python3 ~/AstroDA/xray-pol/scripts/filter_background.py
./event_l2/ixpe03001101_det1_evt2_v01.fits
./event_l1/ixpe03001101_det1_evt1_v01.fits --output rej
```

- I repeated the above step for detectors 2 and 3.

- This produced three background-rejected event files in the `/event_l2` folder with `_rej` appended to the existing filename.

Note: Other scripts can be obtained from the Contributed IXPE Software page.

Defining selection region

Bobrikova *et al.* (2024) used a circular region of radius $100''$ with GX 13+1 at the centroid position. To do so in a similar manner here, we need to open the filtered `_rej` event files on DS9.

Problem: However, when I ran the command
`ds9 ./event_l2/ixpe03001101_det1_evt2_v01_rej.fits &` I encountered the following error:

```
Error in startup script: can't find package base64
while executing
"package require base64"
(file "/usr/share/saods9/library/ds9.tcl" line 216)
```

Fix: We can bypass this issue by telling the Terminal to ignore HEASoft's library paths just for the brief moment it launches DS9. To do so, I modified the above command as follows:
`env -u LD_LIBRARY_PATH ds9 ./event_l2/ixpe03001101_det1_evt2_v01_rej.fits &`

Data files for analysis

The I, Q and U PHA files were extracted using `Xselect` as described in the journal entry dated 23 Sep. 2025 (Tuesday).

The ARF/MRF and RMF files were created using the `ixpecalcarf` and `quzcif` tasks respectively.

Jun. 16, 2026 — Tuesday

- Continuation of the walkthrough follows.

Association of file headers

The various files were renamed using a simpler convention as follows:

- The source spectra files are renamed as, for example for detector 1:
  `d1_I.src d1_Q.src d1_U.src`
- The background spectra files are renamed as, for example for detector 1:
  `d1_I.bkg d1_Q.bkg d1_U.bkg`
- The ARF file is renamed as, for example for detector 1: `d1.arf`
- The MRF file is renamed as, for example for detector 1: `d1.mrf`
- A soft link is created for the appropriate RMF file as, for example for detector 1:

  ```
  d1.rmf -> ixpe_d1_20211209_alpha075_01.rmf
  ```

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- Continuation of the walkthrough follows.

Association of file headers

The various files are associated with the source spectrum file as follows.

Detector 1:

```
fthedit d1_I.src \
> keyword=RESPFILE \
> operation=add \
> value='d1.rmf' \
> longstring=YES
```

```
fthedit d1_Q.src \
> keyword=RESPFILE \
> operation=add \
> value='d1.rmf' \
> longstring=YES
```

```
fthedit d1_U.src \
> keyword=RESPFILE \
> operation=add \
> value='d1.rmf' \
> longstring=YES
```

```
fthedit d1_I.src \
> keyword=ANCRFILE \
> operation=add \
> value='d1.arf' \
> longstring=YES
```

```
fthedit d1_Q.src \
> keyword=ANCRFILE \
> operation=add \
> value='d1.mrf' \
> longstring=YES
```

```
fthedit d1_U.src \
> keyword=ANCRFILE \
> operation=add \
> value='d1.mrf' \
> longstring=YES
```

Detector 2:

```
fthedit d2_I.src \
> keyword=RESPFILE \
> operation=add \
> value='d2.rmf' \
> longstring=YES
```

```
fthedit d2_Q.src \
> keyword=RESPFILE \
> operation=add \
> value='d2.rmf' \
> longstring=YES
```

```
fthedit d2_U.src \
> keyword=RESPFILE \
> operation=add \
> value='d2.rmf' \
> longstring=YES
```

```
fthedit d2_I.src \
> keyword=ANCRFILE \
> operation=add \
> value='d2.arf' \
> longstring=YES
```

```
fthedit d2_Q.src \
> keyword=ANCRFILE \
> operation=add \
> value='d2.mrf' \
> longstring=YES
```

```
fthedit d2_U.src \
> keyword=ANCRFILE \
> operation=add \
> value='d2.mrf' \
> longstring=YES
```

Detector 3:

```
fthedit d3_I.src \
> keyword=RESPFILE \
> operation=add \
> value='d3.rmf' \
> longstring=YES
```

```
fthedit d3_Q.src \
> keyword=RESPFILE \
> operation=add \
> value='d3.rmf' \
> longstring=YES
```

```
fthedit d3_U.src \
> keyword=RESPFILE \
> operation=add \
> value='d3.rmf' \
> longstring=YES
```

```
fthedit d3_I.src \
> keyword=ANCRFILE \
> operation=add \
> value='d3.arf' \
> longstring=YES
```

```
fthedit d3_Q.src \
> keyword=ANCRFILE \
> operation=add \
> value='d3.mrf' \
> longstring=YES
```

```
fthedit d3_U.src \
> keyword=ANCRFILE \
> operation=add \
> value='d3.mrf' \
> longstring=YES
```

Analyzing data in Xspec

I. Loading into data and plot groups

```
XSPEC12>data 1:1 d1_I.src
XSPEC12>data 1:2 d1_Q.src
XSPEC12>data 1:3 d1_U.src

XSPEC12>data 2:4 d2_I.src
XSPEC12>data 2:5 d2_Q.src
XSPEC12>data 2:6 d2_U.src

XSPEC12>data 3:7 d3_I.src
XSPEC12>data 3:8 d3_Q.src
XSPEC12>data 3:9 d3_U.src
```

II. Plotting the I spectra alone

```
XSPEC12>plot 1 4 7 data
```

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Roadmap for replicating the results of Bobrikova *et al.* (2024):

1. Setup of the data and environment

Task	Status
• To download IXPE data for ObsID 03001101 (25 Feb. 2024)	Complete
• To install `ixpeobssim` and HEASoft	Complete
• To update CALDB to the version released on 28 Feb. 2024	Complete

2. Pre-processing of the IXPE data

Task	Status
• To create a source extraction region of 100” centred at GX 13+1	Complete
• To refrain from background photon extraction because GX 13+1 is a very bright source	
• To focus analysis on the 2–8 keV energy band	

3. Generation of lightcurves and hardness analysis

Task	Status
• To use the `xpbin` tool with the `LC` algorithm by grouping the three IXPE DUs and using 200 s time bins	
• To calculate the ratio of counts in the 4–8 keV band to the 2–4 keV band over time to replicate the bottom panel of Fig. 1	

4. Model-independent analysis: using PCUBE algorithm in `xpbin` to calculate the average polarization

Task	Status
• To split the 90 ks observation into five equal 9.5-hour bins to check for variability in the normalized Stokes parameters (q and u)	
• To bin the data into energy intervals (e.g., 2–3 keV, 3–4 keV, 4–5.5 keV, and 5.5–8 keV) to see the polarization's energy dependence	

5. Model-dependent analysis: spectro-polarimetric fitting using XSPEC

Task	Status
• To perform a joint spectral fit with Swift/XRT data using the model `tbabs*(diskbb+bbodyrad)`	
• To apply the `polconst` model to the entire continuum to find the average values	
• To use the `pollin` model to estimate how polarization degree (PD) and angle (PA) change with energy	